
Reliable Causal Estimation under Latent Markov Confounding

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Abstract

When a persistent latent regime drives both a treatment and an outcome, a standard remedy is to infer the regime with a hidden Markov model and condition on its posterior. We show that this soft conditioning leaves a bias with a closed form, an omitted-variable-bias identity in the residual state variance. The bias is governed by the *residual state variance* v left in the posterior and by two regime shifts: one in the treatment (Δ_T) and one in the outcome (Δ_g). It equals $\Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2)$, where σ^2 is the exogenous treatment variance. The bias is non-monotone in the treatment shift Δ_T (it is linear in Δ_g), so the conditioning of the residualized treatment (the usual Gram-matrix $\lambda_{\min}$) is not a monotone predictor of reliability. The quantity v is estimable by a plug-in for calibrated posteriors, with error at most $\sqrt{CE} + CE$ in the calibration error CE (which itself requires labels or a validation set; under misspecified emissions the plug-in is only a lower bound), leaving Δ_g as the single unidentified parameter, which yields a one-parameter sensitivity analysis and a robustness value. In learned-HMM simulations with dependent data, linear and boosted nuisances, and two or three regimes, the plug-in law (with Δ_T, σ^2 known) predicts the realized bias with a median relative error of about 5% for two regimes (7% over all 72 cells; 14% when $\hat{a}$ and $\text{Var}(\hat{T})$ are also estimated). For three regimes it tracks the large biases but under-predicts small ones. For EM-fitted parameters a rate is derived under explicit hypotheses we do not verify for EM (App. A). It also reproduces the bias of a difficulty-frontier benchmark, and it shows that a regime-aware

coupled estimator is no better than regime fixed effects. On four Indian air-quality networks, the Delhi estimate is robust to residual confounding through the HMM-regime channel, provided the posterior is calibrated and no other confounder acts; the other cities are inconclusive.

1 INTRODUCTION

Observational time series are often shaped by a persistent hidden regime: a boundary-layer state in atmospheric data, a macro regime in finance, an operating mode in a plant. If the regime moves both the treatment and the outcome, adjusting for observed covariates X_t does not identify the effect. A common remedy is to fit a hidden Markov model (HMM) to auxiliary proxies Z_t and then condition on the regime posterior $\hat{\gamma}_t$, either as a hard label or as a soft probability, inside a double machine learning (DML) estimator (Chernozhukov et al., 2018). The practical question is how far the result can be trusted when the posterior is imperfect.

Two quantities are commonly used to answer it: the uncertainty of the posterior (its entropy or L^1 error ε_γ) and the conditioning of the score Jacobian, whose smallest eigenvalue $\lambda_{\min}$ measures how much treatment variation is left after residualization. Heuristically, error then scales like $\varepsilon_\gamma / \lambda_{\min}$. We show that this heuristic is wrong in a specific, checkable way, and we replace it with a closed-form law (exact for the learner-defined residual variance, Remark 1).

Contributions. (1) *A residual-state bias law* (Thm. 1, Cor. 1). For soft conditioning on a posterior computed from the proxy sequence, $\hat{\theta} - \theta = \Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2) + O_P(N^{-1/2})$, with a matrix version for $K > 2$ regimes. This is proved for a posterior that is a fixed function of the proxies; for EM-fitted parameters it holds at rate $N^{-1/2}$ under hypotheses H1–H3 (App. A), which we state and test numerically but do not verify for EM. (2) *Non-monotone difficulty*. The bias is hump-shaped in Δ_T with peak $|\Delta_g| \sqrt{v} / (2\sigma)$, whereas the residual treat-

ment variance (the scalar analogue of $\lambda_{\min}(\hat{J})$) rises monotonically. Refining the conditioning information can only reduce the bias (Prop. 1); soft posteriors beat hard regime labels. (3) *Observability and sensitivity analysis* (Prop. 2, Sec. 4). v is estimated by the posterior Gini impurity, with error at most $\sqrt{CE} + CE$ in the posterior’s calibration error (CE requires the true state or a labelled validation set, so on real data calibration must be argued, not measured). Only Δ_g is unidentified, giving a robustness value that says how large a hidden outcome shift must be to explain an estimate away. (4) *Evidence with strong baselines* (Sec. 7). The plug-in law matches simulations to a median relative error of about 5% ($K=2$, cells with $|\text{bias}| > 0.05$, Δ_T, σ^2 known; $K=3$: tracks the two large-bias cells and under-predicts the rest, median error 43% over 8 cells and 10% with the realized v_N), explains the bias of our own difficulty benchmark, exposes an experimental pitfall (nuisances that omit the regime), and is applied to four Indian cities.

Relation to prior work. Mathematically the law is an omitted-variable-bias (OVV) identity with a partially observed confounder (Greenland, 1980; Cinelli and Hazlett, 2020; Chernozhukov et al., 2022); the hump follows from the same algebra that underlies bias amplification (Pearl, 2010; Ding et al., 2017). We do not claim these ingredients as new. What we add is (i) the identification of the residual variance with the calibrated HMM posterior, valid for learned posteriors and dependent data under purged cross-fitting; (ii) the consequence that the Gram-matrix conditioning, a standard DML diagnostic, is not a monotone predictor of reliability here; and (iii) a calibration-based, nearly fully observable sensitivity analysis (whose plug-in $\hat{v}$ is a lower bound under emission misspecification) for latent Markov regimes. Posterior-based adjustment is known to fail without identification (Wang and Blei, 2019; D’Amour, 2019); our law quantifies how it fails. Latent-state and proximal approaches (Miao et al., 2018; Tchetgen Tchetgen et al., 2020) target identification, which we do not assume. HMMs follow Hamilton (1989); Cappé et al. (2005); dynamic DML is in Lewis and Syrgkanis (2020).

2 SETTING AND ESTIMATOR

Let $S_t \in \{0, \dots, K-1\}$ be a latent persistent regime with indicator vector $D_t = (\mathbb{1}\{S_t = k\})_{k=1}^{K-1}$. We observe a treatment T_t , an outcome Y_t , controls X_t , and proxies Z_t that depend on S_t . We posit

$$T_t = m(X_t) + a^\top D_t + V_t, \quad (1)$$

$$Y_t = \theta T_t + g(X_t) + b^\top D_t + U_t, \quad (2)$$

where $a \in \mathbb{R}^{K-1}$ and $b \in \mathbb{R}^{K-1}$ are the regime shifts of the treatment and the outcome (Δ_T and Δ_g when $K = 2$), θ is the causal effect, and (V_t, U_t) are exogenous noises (assumptions below). The regime is confounding when $a \neq 0$ and $b \neq 0$.

Soft conditioning. Let $\hat{\gamma}_t = \hat{\gamma}_t(Z_{1:N}) \in \Delta^{K-1}$ be any posterior computed from the proxies, for example the smoothing or filtering distribution of a Gaussian HMM whose parameters are learned by EM on the same sample. With features $F_t = (X_t, \hat{\gamma}_t)$ and purged block cross-fitting with an embargo of τ steps, let $\tilde{T}_t = T_t - \hat{m}_T(F_t)$ and $\tilde{Y}_t = Y_t - \hat{m}_Y(F_t)$ be cross-fitted residuals and

$$\hat{\theta} = \sum_t \tilde{T}_t \tilde{Y}_t / \sum_t \tilde{T}_t^2. \quad (3)$$

Hard assignment uses $\mathbb{K}\{\arg \max \hat{\gamma}_t\}$ instead of $\hat{\gamma}_t$. The coupled, regime-specific estimator of our earlier work (OR-DML) is analysed in Appendix E; it does not improve on these baselines.

3 THE RESIDUAL-STATE BIAS LAW

Let $c_t = \mathbb{E}[D_t | F_t]$, $\tilde{D}_t = D_t - c_t$, $\Sigma_N = N^{-1} \sum_t \tilde{D}_t \tilde{D}_t^\top$ (the residual state covariance), and $v = \Sigma_N$ when $K = 2$.

Assumption 1 (Exogenous noise). V_t is independent of (S, Z, X) at all times, $\mathbb{E}V_t = 0$, $\text{Var}V_t = \sigma^2 > 0$. $\mathbb{E}[U_t | S, Z, X, V] = 0$. Both noise processes have summable autocovariances and bounded $4+\delta$ moments.

Assumption 2 (Posterior as a local feature). The process is stationary and α -mixing. $\hat{\gamma}_t$ is a function of $Z_{1:N}$ that depends on Z_s for $|s - t| > \tau$ only through terms of size ε_τ in L^2 (filter stability), and estimated HMM parameters converge at rate $N^{-1/2}$.

Assumption 3 (Nuisance learners). The cross-fitted learners for $\mathbb{E}[T | F]$ and $\mathbb{E}[Y | F]$ have L^2 errors r_T, r_Y (set $r_h = r_Y + |\theta|r_T$) and the embargo τ exceeds the mixing time.

Theorem 1 (Residual-state bias law). *Under Assumptions 1–3,*

$$\hat{\theta} - \theta = \frac{a^\top \Sigma_N b}{a^\top \Sigma_N a + \sigma^2} + O_P(N^{-1/2} + r_T r_h + \varepsilon_\tau), \quad (4)$$

with constants depending on $\|a\|, \|b\|$ and σ^{-2} . In particular, if the remainder vanishes, $\hat{\theta}$ is consistent iff $a^\top \Sigma_N b \rightarrow 0$, and the denominator equals $\text{Var}(\tilde{T})$.

Remark 1 (v is relative to the learner). Theorem 1 is stated for the unrestricted conditional covariance Σ_N . With a restricted learner class the relevant Σ_N is the residual of the learner’s projection, which can exceed $\mathbb{E}\text{Var}(S | F)$, and exactness additionally needs m, g

in the class. We observe both effects: with boosted nuisances the law evaluated at the realized v_N has 14% median error (versus 1–3% for linear learners), and in the calibration experiment it overshoots by 26%, because the cross-fitted learner adds estimation noise to v_N (Limitation 5, Sec. 7).

The estimation error of the HMM enters *only through* v , because v is defined with respect to the realized posterior. The proof (Appendix A) writes the residuals as $\tilde{T} = a^\top \tilde{D} + V - \delta^T$ and $\tilde{Y} - \theta \tilde{T} = b^\top \tilde{D} + U - \delta^h$; all cross terms are centered mixing sums or nuisance products.

Intuition. After partialling out F_t , the treatment still contains the part of the regime that the posterior failed to explain, $a^\top \tilde{D}_t$, and so does the outcome, $b^\top \tilde{D}_t$. This shared component is a confounder that no longer has a proxy: it contributes covariance $a^\top \Sigma_N b$ to the numerator, while the denominator mixes it with the clean variation σ^2 . A strong treatment shift a makes the unexplained component dominate the treatment residual (so $\text{Var}(\tilde{T})$ is large), but then the estimator regresses $\tilde{Y}$ on a variable that is nearly a copy of $b^\top \tilde{D}$, and its slope tends to $\theta + b/a$; the bias is free of θ and shrinks like b/a as a grows. Weak shifts leave little bias to begin with.

Corollary 1 (Hump, sup bound, information monotonicity). *Let $K = 2$ and $B = \Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2)$ with $\lambda := \Delta_T^2 v + \sigma^2 = \text{Var}(\tilde{T})$. (i) $|B| \leq |\Delta_g| \sqrt{v} / (2\sigma)$ for all Δ_T , with equality at $|\Delta_T| = \sigma / \sqrt{v}$. (ii) $|B| = |\Delta_g| \sqrt{v} \sqrt{\lambda - \sigma^2} / \lambda$, which increases in λ on $[\sigma^2, 2\sigma^2]$ and decreases afterwards, while λ increases monotonically in $|\Delta_T|$. (iii) For fixed $(\Delta_T, \Delta_g, \sigma)$, $|B|$ is strictly increasing in v .*

Part (ii) is the correct replacement for the heuristic $\varepsilon_\gamma / \lambda_{\min}$: the numerator $\Delta_T \Delta_g v$ grows with the same confounding strength that inflates λ , so no function of $(\varepsilon_\gamma, \lambda)$ alone bounds the bias monotonically. The bound (i) decays only as $\sqrt{v}$, so the worst case is $O(\sqrt{\varepsilon_\gamma})$ even though for fixed Δ_T the bias is linear in v .

Proposition 1 (Information monotonicity). *If $\mathcal{F} \subseteq \mathcal{F}'$ are two conditioning sets, then $\Sigma' \preceq \Sigma$, and for $K = 2$ $|B'| \leq |B|$. In particular, conditioning on the soft posterior never does worse than conditioning on its hard label, and adding informative controls or proxies cannot hurt.*

Proposition 2 (Observable plug-in). *Let $\hat{\Sigma} = N^{-1} \sum_t [\text{diag}(\hat{\gamma}_t) - \hat{\gamma}_t \hat{\gamma}_t^\top]_{1,1}$. (for $K = 2$, $\hat{v} = N^{-1} \sum_t \hat{\gamma}_t (1 - \hat{\gamma}_t)$, the posterior Gini impurity) and $CE = \mathbb{E} \|\hat{\gamma}_t - \mathbb{E}[D_t | F_t]\|^2$ the L^2 calibration error given F_t . For $K = 2$,*

$$|\mathbb{E} \hat{v} - \mathbb{E} v| \leq \sqrt{CE} + CE, \quad (5)$$

so $\hat{v} = v$ for a calibrated posterior and an over-confident

Algorithm 1 Residual-state sensitivity analysis

- 1: Fit an HMM to $Z_{1:N}$; where labels or a validation set permit, check calibration and recalibrate $\hat{\gamma}$ (Prop. 2); otherwise treat $\hat{v}$ as a lower bound on v (over-confident posterior or misspecified emissions: factor 2 in `x29`).
 - 2: Cross-fit $\hat{m}_T, \hat{m}_Y$ on $(X, \hat{\gamma})$ with purged blocks; compute $\tilde{T}, \tilde{Y}, \hat{\theta}$.
 - 3: Compute $\hat{\Sigma}$ (posterior Gini), $\hat{a}$ (OLS of T on $(1, X, \hat{\gamma})$), $\widehat{\text{Var}}(\tilde{T})$.
 - 4: For a grid of outcome shifts b : report $\hat{\theta}_{\text{adj}}(b)$; report b^* in SD of $\tilde{Y}$ and its bootstrap interval.
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one understates the bias (so does a misspecified emission model, App. A, `x29`). If m is linear and the posterior is calibrated, the OLS coefficient $\hat{a}$ of $\hat{\gamma}$ in the regression of T on $(1, X, \hat{\gamma})$ consistently estimates a , and $\sigma^2 = \text{Var}(\tilde{T}) - a^\top \Sigma a$.

Heterogeneous effects $\theta(S)$ add the terms $a(\theta_1 - \bar{\theta}) + \bar{m}(\theta_1 - \theta_0)$ to b (Appendix B), where $\bar{m}$ is the state-free mean treatment level.

4 SENSITIVITY ANALYSIS

By Thm. 1 and Prop. 2, everything in the bias except b is observable: $\hat{a}$ from the treatment regression, $\hat{\Sigma}$ from the posterior, and $\widehat{\text{Var}}(\tilde{T})$ from the residuals. For a hypothesized outcome shift b the bias-adjusted estimate is $\hat{\theta}_{\text{adj}}(b) = \hat{\theta} - \hat{a}^\top \hat{\Sigma} b / \widehat{\text{Var}}(\tilde{T})$. **Robustness value.** The smallest outcome shift that explains the point estimate away is

$$b^* = \frac{|\hat{\theta}| \widehat{\text{Var}}(\tilde{T})}{\|\hat{\Sigma} \hat{a}\|}, \quad (6)$$

reported in standard deviations of $\tilde{Y}$ and benchmarked against the observed regime shift of Y (the coefficient of $\hat{\gamma}$ in an outcome regression), which is not identified but is a natural yardstick. A large b^* means the estimate cannot be overturned by residual regime confounding through this channel; b^* is uninformative when $\hat{\theta} \approx 0$. A block bootstrap (blocks of 72 rows, 300 draws; percentile intervals) resamples contiguous blocks of rows and recomputes $(\hat{\theta}, \hat{v}, \hat{a}, \widehat{\text{Var}}(\tilde{T}))$.

5 A PRIORI BIAS FROM SENSOR PRIMITIVES

The law takes v as given. For a continuous latent confounder, v has a closed form in the sensor’s information rate and the state’s persistence, so the bias of posterior-conditioned DML can be predicted before (T, Y) are seen.

Proposition 3 (Wiener smoothing error). *Let u_t be a stationary unit-variance $AR(1)$ with coefficient ϕ , observed as $Z_t = hu_t + \varepsilon_t$ with $\varepsilon_t \sim N(0, R)$ i.i.d., and let $\iota = h^\top R^{-1}h$. Conditioning on the Kalman-smoothed mean leaves residual variance*

$$v = \frac{1 - \phi^2}{\sqrt{(1 + \phi^2 + \iota(1 - \phi^2))^2 - 4\phi^2}}, \quad (7)$$

and the bias is $\Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2)$ by Thm. 1 with u in place of S . For $0.02 \ll \iota \ll (1 - \phi^2)^{-1}$, $v \approx \frac{1}{2} \sqrt{(1 - \phi^2)/\iota}$, so the worst-case bias $|\Delta_g| \sqrt{v}/(2\sigma)$ decays only as $\iota^{-1/4}$; for $\iota \gg (1 - \phi^2)^{-1}$, $v \approx 1/\iota$ and it decays as $\iota^{-1/2}$.

The proof is the Wiener–Kolmogorov smoothing error $v = \int S_u / (1 + \iota S_u) d\omega / 2\pi$ with $S_u(\omega) = (1 - \phi^2) / |1 - \phi e^{-i\omega}|^2$, evaluated with $\int_0^{2\pi} d\omega / 2\pi (A - 2\phi \cos \omega)^{-1} = (A^2 - 4\phi^2)^{-1/2}$ (Anderson and Moore, 1979). In 18 simulated configurations ($\phi \in \{0.9, 0.97\}$, three sensor strengths, three Δ_T ; $N=3000$, 8 replications each; Table 1), the closed form matches the Kalman smoother’s variance to 2×10^{-4} , and the predicted bias matches the DML bias with correlation 0.9987 and maximum error 0.021. For a persistent state, halving the worst-case bias therefore needs about $16\times$ the sensor information while $\iota \ll (1 - \phi^2)^{-1}$. Discretizing u into a K -state HMM posterior acts as a sieve: at $\phi = 0.97$, $\iota = 2.76$, $\Delta_T = 1$ (6 replications, $N=3000$), the bias falls from 0.82 ($K=2$) to 0.52 ($K=3$) and 0.39 ($K=4$) and then stops improving (0.40 at $K=6$), matching the law with $v_K = \text{Var}(u | X, \hat{\gamma}_K)$ (0.78, 0.49, 0.37, 0.38).

Finite-state chains. For a finite-state chain, $v = \mathbb{E}\text{Var}(S | Z_{1:N})$ is the minimum mean-squared smoothing error, so it is at most the linear Wiener error of the rescaled indicator: $v \leq \frac{1}{4} v_W(2\rho - 1, \iota)$ with $\iota = \Delta_Z^2 h^\top R^{-1}h$ for emission means $\pm \Delta_Z h$ (a bound, valid for known parameters). It is tight for weak signals (bound/exact ratio 1.04–1.4 for $\Delta_Z \leq 0.5$ at $\rho = 0.9$, and 1.3–2.3 at $\rho = 0.97$; `x8_finite_state.csv`) and loose for well-separated regimes, where v falls exponentially (exact $v = 10^{-4}$ vs. bound 0.017 at $\Delta_Z = 2$; App. C). Persistent discrete regimes with informative proxies are therefore far less dangerous than a continuous latent state with the same sensor information, whose bias decays only polynomially.

Change-point scaling. For a persistent two-state chain with Gaussian emissions and Mahalanobis separation $d^2 = (\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)$, residual variance is paid at change points. Let $r = \pi_0 \epsilon_0 + \pi_1 \epsilon_1$ be the switch rate per step. In the isolated-change limit, the posterior of the change location c is $p(k) \propto e^{W_k}$, where $W_{\pm 1} \sim N(-d^2/2, d^2)$ is the log-likelihood ratio of moving the change by one site, and only the two sites adjacent to

the switch carry non-negligible variance, each contributing $\mathbb{E}[\sigma(W)\sigma(-W)] = \frac{e^{-d^2/8}}{d\sqrt{2\pi}} \int \frac{e^{-w/2} e^{-w^2/2d^2}}{4 \cosh^2(w/2)} dw \rightarrow \frac{\pi}{2} \frac{e^{-d^2/8}}{d\sqrt{2\pi}}$ (the odd part of $e^{-w/2}$ integrates out and $\int \frac{dw}{4 \cosh^2(w/2)} = \frac{\pi}{2}$). Hence $v \approx \sqrt{\pi/2} r e^{-d^2/8}/d$ as $r \rightarrow 0$ and $d \rightarrow \infty$. The nearest-neighbour term is exact (quadrature, `x20_nn_quad.csv`): its constant tends to $\sqrt{\pi/2}$ with relative gap $\approx \pi^2/(2d^2)$ (an asymptotic expansion: the first-order truncation is inaccurate below $d \approx 6$); the quadrature values are 0.91, 1.01, 1.08, 1.12, 1.17 at $d = 3, 4, 5, 6, 8$, within 0.3% of the limit at $d = 40$. That the limit is approached only as d^{-2} is why simulations at $d \geq 4$ sit below 1.25 (at $d \leq 3$ the total exceeds it). The remaining sites add +88%, +30%, +9.7%, +3.2%, +0.8% at $d = 2, \dots, 6$ (importance-sampled total constant for one isolated change relative to the exact nearest-neighbour quadrature, `x28_cp_total.csv`; the total itself is 1.39, 1.18, 1.11, 1.11, 1.13, standard error ≤ 0.015 ; the $d = 8$ estimate is too noisy to use and $d \geq 10$ failed through weight degeneracy, so we state nothing there). These are measured, not bounded, and the interaction of consecutive switches is not controlled, so the full statement is a derivation sketch, not a proof. With the exact HMM posterior on windows containing one switch (variance-reduced, $\epsilon = 10^{-4}$; `x19_hmm_window.csv`) the constant $v d e^{d^2/8}/\epsilon$ is 1.19 ($d = 3$), 1.12 ($d = 4$) and 1.12 ($d = 5$), standard error ≤ 0.004 (1.21, 1.14, 1.14 at $\epsilon = 10^{-3}$); our earlier stationary-chain estimate of 1.23 at $d = 4$ was Monte Carlo noise (about 1500 switches, rare-event tail) and is superseded, and for $\epsilon \geq 0.03$ the constant is larger (1.5–2.7 for $d \geq 3$), which is why the practical formula uses $\sqrt{8/\pi}$ (App. C). For the practical range $\rho \in [0.9, 0.97]$ an effective constant $\sqrt{8/\pi}$ fits:

$$v \approx \sqrt{8/\pi} r \frac{e^{-d^2/8}}{d}. \quad (8)$$

Against the exact HMM filter (40,000 steps, $\rho \in \{0.8, \dots, 0.99\}$) this is within $\pm 10\%$ for $\rho \in \{0.9, 0.95\}$ and $\pm 14\%$ at $\rho = 0.97$ (up to +32% at $\rho = 0.8$) ($\Delta_Z \in [0.5, 1.5]$, $d^2 \in [2.8, 25]$), within 0.89–1.11 for a one-dimensional proxy at equal d^2 (`x17_oned_check.csv`), and within 0.87–1.02 for asymmetric chains with stationary regime probability as low as 0.1 (so the relevant quantity is the switch rate r , not ρ); it drifts up to +23% at $\rho = 0.99$ and is noise-dominated at $\Delta_Z = 2$ where $v < 10^{-3}$. Plugging this *a priori* v into the law predicts the learned-HMM bias of Sec. 7 from sensor primitives alone, but the constant was fitted on exact-HMM v from the same generator, so this is a consistency check, not an out-of-sample test. RMSE over the cells with $\Delta_Z \geq 0.8$: smoother with linear nuisances 0.0059 (data-based $\hat{v}$: 0.0080), smoother with boosted nuisances 0.0155 (0.0115), filter with linear nuisances 0.0362 (0.0121); the asymptotic constant $\sqrt{\pi/2}$

gives 0.0150, 0.0111, 0.0504 (`x18.apriori.check.csv`). The a priori law is thus competitive only for smoothing posteriors with linear nuisances; e.g. $\Delta_Z=0.8$, $\Delta_T=8$: predicted 0.230, observed 0.226. Follow-up checks show the form carries over but the constants do not: a three-state chain (nearest-pair d^2) gave $0.74\text{--}0.84\times$ the two-state value of $\sum_k \gamma_k(1-\gamma_k)$, and unit-variance t_5 proxies (true density in the filter) gave $0.9\text{--}1.7\times$, increasing with $d^2 \in [6, 17]$ (`x13.switchrate.ext.csv`). The one-change simulation (`x15`) is consistent with the nearest-neighbour quadrature (e.g. 1.02 vs. 1.01 at $d = 4$).

Table 1: Bias predicted from primitives vs. observed ($\Delta_g=3$, $\Delta_T=1$, $\sigma^2=1$).

ϕ	ι	closed-form v	predicted bias	observed bias
0.9	0.69	0.261	0.622	0.601
0.9	2.76	0.128	0.340	0.337
0.9	11.04	0.055	0.156	0.154
0.97	0.69	0.146	0.383	0.373
0.97	2.76	0.073	0.203	0.219
0.97	11.04	0.034	0.100	0.107

6 BEYOND SENSITIVITY: WHEN CAN b BE LEARNED?

The law leaves b unidentified. Two routes learn it, and they fail differently. **Joint model.** Treat (Z, T, Y) as emissions of the chain (a Markov-switching regression with a common θ) and maximize the likelihood by EM. When the regime model is correct this is essentially unbiased and has the smallest variance in our simulations, even when Z is uninformative (App. C); the price of the modular, two-stage posterior used above is exactly the law. But it inherits the regime model: if the latent confounder is a continuous AR(1) process, a $K=2$ joint fit is as biased as posterior-DML (about $+0.8$ on $\theta=1$). **Proximal route.** Suppose a negative-control outcome $W = cS + \ell T + e$ is available ($c \neq 0$; $\ell = 0$ means T does not cause W), and $Z \perp (U, V, e) \mid S$.

Proposition 4 (Proximal identification, linear case). *Under the linear model of Sec. 2, the 2SLS regression of $\tilde{Y}$ on $(\tilde{T}, \tilde{W})$ with instruments $(\tilde{T}, \tilde{\gamma})$, where tildes denote cross-fitted residuals on X , converges to $(\theta - \ell b/c, b/c)$. Hence $\ell = 0$ identifies θ and b/c (so b if c is known) without any model for the latent state, and for $\ell \neq 0$ the bias is $-\ell \hat{\beta}_W$, computable for any hypothesized leakage ℓ .*

This is the linear special case of proximal causal inference (Miao et al., 2018; Tchetgen Tchetgen et al., 2020); we claim no novelty for the identification argument. What the regime setting adds is (i) the posterior $\hat{\gamma}$ is a valid instrument when W is an external

variable, but *not* when W is a lagged outcome, in our checks (see below) and (ii) in a continuous-latent stress test, proximal 2SLS has bias below 0.03 where posterior-DML and the joint MLE are off by 0.8–0.9 (Table 7). The law still describes the failure: replacing S by the continuous state u and v by $\text{Var}(u \mid X, \hat{\gamma})$ predicts the bias to within 1.4–3.2%, slightly under: 0.808 vs. 0.834 and 0.750 vs. 0.774 at $\Delta_T=1$ (sensor SNR 0.5, 1), 0.888 vs. 0.901 and 0.861 vs. 0.874 at $\Delta_T=2$ (`x3b.contlat.law.csv`). The cost is variance (RMSE 0.02–0.16 vs. 0.01–0.03 for the correct joint model) and reliance on a valid negative control; a leaking control ($\ell = 0.3$) gives bias $-0.45 = -\ell b/c$, as predicted. We also tried the lagged outcome Y_{t-h} as a self-contained negative control ($N=20,000$, $h=4$, $\Delta_T=1$, 4 replications; `x4c.selfnc.check.csv`). With iid shocks, proximal 2SLS with the posterior as instrument has mean bias 0.094 and with raw Z as instrument 0.018, versus 0.111 for DML; with AR shocks they are 0.148 and 0.098 versus 0.110. Neither instrument is reliable, so we do not recommend this variant, and we do not claim a mechanism. On the four cities we do not apply the proximal estimator: the available proxies (meteorology) also affect $\text{PM}_{2.5}$ directly, so the exclusion restrictions fail.

Environment route (no negative control, no joint model). The law is linear in an observable: in environment e , $\theta_e = \theta + b_e k_e$ with $k_e = a_e v_e / \text{Var}(\tilde{T}_e)$, because $\text{Var}(\tilde{T}_e) = a_e^2 v_e + \sigma_e^2$. If the outcome shift is invariant across environments (or merely independent of k_e) while the treatment shift varies, regressing $\hat{\theta}_e$ on $\hat{k}_e$ identifies θ as the intercept, i.e. it extrapolates to zero residual confounding. In a simulation with 10 environments of $N=1500$ and treatment shifts $\Delta_T \in [0.3, 4]$ (12 replications), the intercept has bias $+0.03$ (RMSE 0.04) for a common $b = 3$ and 0.00 (RMSE 0.05) for random b_e , against $+0.30$ for the average of environment-wise estimates; the slope recovers b (3.3, 3.7). If b_e grows with the treatment shift the intercept is biased (-0.14), still smaller than the pooled error ($+0.36$). On the four cities, with calendar months as environments, the route has no leverage: $|k_e| \leq 0.06$ except in Kolkata (4 months), and monthly $\hat{\theta}_e$ is too unstable (standard deviations 0.4–28) for the regression to be informative, so we report no real-data estimate from it (`x11.envreal.csv`).

7 EXPERIMENTS

Setup. All experiments use `src/bias.law/` and share these ingredients unless noted: a persistent Markov chain (stay probability 0.9), 2-d Gaussian proxies with separation Δ_Z , two exogenous controls, $N = 3000$, unit-variance noises, $\Delta_g = 3$. The HMM is fitted by

Table 2: Agreement between measured bias and $\text{law}(\hat{v})$. Median relative error is over cells with $|\text{bias}| > 0.05$ (54 of 72 cells for $K=2$; 2 of 8 for $K=3$, where the all-cell median is 43%).

Setting	cells	$\text{corr}(\hat{\theta} - \theta, \text{law}(\hat{v}))$	RMSE	med. rel. err.	$\text{corr}(\text{law}(v_N))$
$K=2$, filter, lin	24	0.997	0.016	7.3%	1.000
$K=2$, smooth, gbm	24	0.998	0.011	4.3%	0.995
$K=2$, smooth, lin	24	0.998	0.009	5.1%	0.999
$K=3$, smooth, lin	8	0.999	0.010	9.8%	0.998

EM on the same sample (smoother or filter), cross-fitting uses 5 purged blocks with $\tau = 24$, and nuisances are linear or gradient boosting. Reported biases are means over 12 replications per cell with $\theta = 1$; the law is evaluated with the *observable* plug-ins unless stated.

7.1 The law predicts the bias

Figure 1 and Table 2 summarize 72 cells for $K = 2$ (smoother and filter; linear and boosted nuisances; three proxy qualities $\Delta_Z \in \{0.5, 0.8, 1.2\}$; eight values of Δ_T from 0.25 to 32) and 8 cells for $K = 3$ (matrix law, four (a, b) configurations including sign changes). The law with the observable plug-in $\hat{v}$ has correlation 0.996 and RMSE 0.012 with the measured bias, with median relative error 5.2% on cells with $|\text{bias}| > 0.05$. Using the realized v_N gives 3.5%, and also estimating $\hat{a}$ and $\widehat{\text{Var}}(\tilde{T})$ from the data gives 14%. The predicted peak location $\sigma/\sqrt{\hat{v}}$ and height $|\Delta_g|\sqrt{\hat{v}}/2\sigma$ match the observed maximum to within the grid (Appendix D).

Soft beats hard. For the smoother with linear nuisances the soft posterior has smaller $|\text{bias}|$ than the hard regime label in all 24 (Δ_Z, Δ_T) cells (23/24 with boosting; median ratio 1.19 where the bias exceeds 0.02), as Prop. 1 predicts. The filter posterior is worse than the smoother’s hard labels in all 24 cells, since it conditions on less information, which the proposition does not cover; DML on X alone has mean absolute bias 0.387 versus 0.125 for the soft smoother posterior (0.114 boosted, 0.157 filter), 0.142 for the hard label and 0.004 for the oracle state (Fig. 1c).

Using the law to correct. Subtracting the plug-in bias $\hat{a}\hat{v}b/\widehat{\text{Var}}(\tilde{T})$ with the true b reduces the per-replication mean absolute error from 0.126 to 0.021 (83%). Misspecifying b by a factor of 0.5 or 2 gives 0.072 and 0.097 (43% and 23% below the uncorrected error). The correction therefore degrades gracefully but is not robust to gross misspecification, which is why we advocate sensitivity analysis rather than point correction.

Table 3: Table-1 DGP (30 replications; $N = 1200$; law uses the realized v_N). θ estimates; “excess” = $\hat{\theta}_{\text{soft}} - \hat{\theta}_{\text{oracle}}$; true pooled $\bar{\theta} = 1.53$.

Δ_Z	DML on X	soft $\hat{\gamma}$	oracle S	excess bias	$\text{law}(v_N)$	$\text{law}(\hat{v})$
0.2	10.02	9.773	1.535	8.238	8.236	7.418
0.5	10.02	7.776	1.535	6.241	6.267	5.655
1.0	10.02	3.796	1.535	2.260	2.299	1.891
1.5	10.02	1.923	1.535	0.388	0.402	0.320
2.0	10.02	1.559	1.535	0.024	0.024	0.016
4.0	10.02	1.535	1.535	0.000	0.000	0.000

7.2 It explains a standard difficulty benchmark

We rerun our earlier difficulty-frontier DGP ($\theta = (0.75, 2.5)$, treatment shift $\Delta_T = 4$, outcome offsets 15/50, i.e. $\Delta_g = 35$), in which “DML on X only” has bias ≈ 8.5 . Table 3 shows that the *excess* bias of soft conditioning over the oracle-state estimator is predicted by the heterogeneous-effects form of the law (with the realized v_N) to within 0.04 at every Δ_Z , from 8.24 at $\Delta_Z = 0.2$ to 0.02 at $\Delta_Z = 2$; the observable plug-in $\hat{v}$ underestimates it by 9–32%. So in this benchmark the transition between failure and success is accounted for by $v(\Delta_Z)$; we found no evidence that a Jacobian-spectrum property is needed to explain it.

Honest baselines. With the proxy used as controls, a regime-fixed-effects baseline matches the coupled OR-DML estimator at $\Delta_Z \geq 2$ (median absolute bias 0.063 vs. 0.072 at $\Delta_Z = 2$, 0.046 vs. 0.046 at $\Delta_Z = 4$) and beats it at $\Delta_Z = 1$ (mean absolute bias 2.64 vs. 4.02); the spectral penalty changes OR-DML by less than 1.5% (Appendix E). The contribution of this paper is therefore the law and the sensitivity analysis, not a new estimator.

7.3 Why $\lambda_{\min}$ misleads

A common experiment varies the treatment shift while holding the representation fixed and reads the error against $\lambda_{\min}(\hat{J})$. Table 4 (12 replications per cell, boosted nuisances, oracle regime weights) shows the pitfall: if nuisances are fit on X only, the regime shift leaks into $\tilde{T}$, $\lambda_{\min}$ grows from 0.54 to 14.4, and the error is hump-shaped (5.9, 14.0, 5.7). If the nuisances are fit on (X, S) , $\lambda_{\min}$ is flat at 0.51 and the error is 0.06–0.10 for every Δ_T . The hump is the one of Cor. 1 with $v = \text{Var}(S)$, whose predicted peak $\sigma/\sqrt{v} \approx 2$ matches the observed peak at 1.5–3. Because $\lambda_{\min}$ moves while the error is flat or falls, $\lambda_{\min}$ does not govern reliability here.

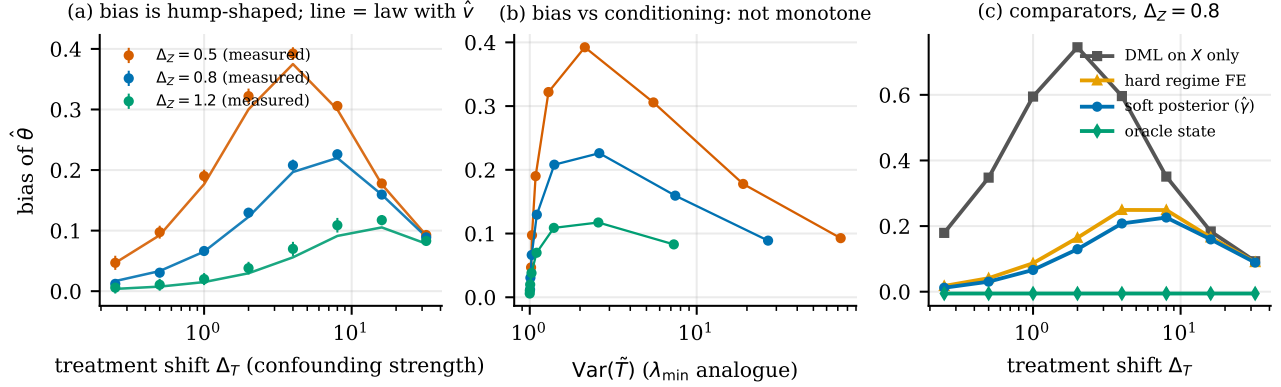


Figure 1: Bias of the soft-conditioning estimator against the treatment shift Δ_T ($K = 2$, learned smoother, linear nuisances, 12 replications, 95% bands). (a) The measured bias is hump-shaped and matches the law (lines) with $\hat{v}$ read from the posterior. (b) The same bias against $\text{Var}(\tilde{T})$, the scalar analogue of $\lambda_{\min}(\hat{J})$: conditioning rises monotonically while bias rises and then falls. (c) Hard regime fixed effects are worse than the soft posterior; DML on X alone is far worse; the oracle state is unbiased.

Table 4: Task-conditioning sweep. Regime-weighted score on the Table-1 DGP, $\Delta_Z = 1.5$.

Δ_T	nuisances on X only $\lambda_{\min}(\hat{J})$	$\ \hat{\theta} - \theta\ _2$	nuisances on (X, S) $\lambda_{\min}(\hat{J})$	$\ \hat{\theta} - \theta\ _2$
0.4	0.54	5.92	0.51	0.10
0.8	0.61	10.34	0.51	0.08
1.5	0.86	14.05	0.51	0.07
3.0	1.88	13.38	0.51	0.07
6.0	5.61	8.76	0.51	0.07
10.0	14.38	5.73	0.51	0.06

Table 5: Temperature scaling ($K = 2$, boosted nuisances, 20 replications, $\Delta_Z = 0.8$, $\Delta_T = 2$, $N = 3000$).

τ	bias	law($\hat{v}$)	law(v_N)	$\hat{v}$	v_N	$\sqrt{CE} + CE$
0.30	0.128	0.033	0.160	0.006	0.030	0.071
0.50	0.127	0.056	0.160	0.010	0.030	0.053
0.75	0.127	0.088	0.160	0.016	0.030	0.035
1.00	0.127	0.123	0.160	0.022	0.030	0.025
1.50	0.127	0.201	0.160	0.039	0.030	0.038
2.50	0.127	0.355	0.160	0.077	0.030	0.098

7.4 Calibration controls the observable bias

We rescale the posterior logits by a temperature τ ($\tau < 1$ over-confident) and compare the law with $\hat{v}$ against the realized residual variance (Table 5). A flexible learner is invariant to monotone recalibration, so the realized bias is constant (0.127) while $\hat{v}$ changes by a factor of 14; the bound $|\hat{v} - v_N| \leq \sqrt{CE} + CE$ held in 100% of the 120 runs, where v_N is the cross-fitted learner residual variance and CE is estimated by in-sample isotonic regression of S on the recalibrated posterior alone (which biases CE downward and ignores X), so this is a weak check of Prop. 2. An over-confident posterior therefore understates the bias, and the sensitivity analysis should use a recalibrated posterior.

7.5 Four Indian cities

We apply the analysis to NO_2 (treatment) and $\text{PM}_{2.5}$ (outcome) on a regular hourly grid in Delhi, Mumbai,

Bengaluru and Kolkata (Feb 2025–Jun 2026; Kolkata ends Jun 2025) with the controls and proxies of Appendix F. Figure 2 and Table 6 report the pooled estimate and the robustness value. (An earlier version of our data pipeline rounded half-hour timestamps to even hours and produced a 2-hourly series; the corrected hourly series changes the numbers materially, e.g. Delhi’s $\hat{\theta}$ from 0.58 to 0.37, Mumbai’s screened estimate from 4.9 to 0.3, Kolkata’s from 0.4 to 4.5; the 2-hourly results are in Table 10, App. F.) For Delhi the NO_2 – $\text{PM}_{2.5}$ association is positive and stable ($\hat{\theta} = 0.37$, 90% interval 0.29–0.44), with a robustness value of about 82 standard deviations of $\tilde{Y}$ (90% interval 54–144), far above the regression coefficient $\hat{c}_Y$ of $\hat{\gamma}$ in the $\text{PM}_{2.5}$ regression ($b^*/|\hat{c}_Y| \approx 560$; a conditional coefficient, not Δ_g , so only a rough yardstick): residual confounding through this channel would need an implausibly large outcome shift to overturn it. The other cities are not informative. Mumbai’s and Bengaluru’s intervals include zero, and Mumbai’s estimate depends on screening runs of ≥ 6 identical NO_2 readings (2.2 unscreened, 0.3 screened, 1,857 rows or 23% removed). Kolkata has only 2,322 hourly rows and its NO_2 takes 16 distinct

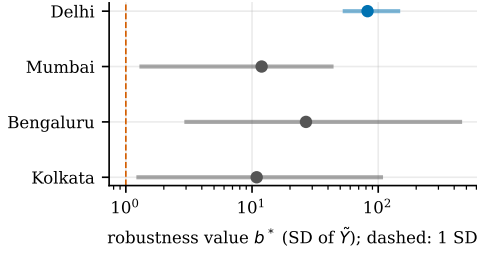


Figure 2: Robustness values b^* (log scale) with 90% block-bootstrap intervals; blue: $\hat{\theta}$ clearly nonzero.

Table 6: Real-data sensitivity (pooled soft-conditioning estimate, Ridge nuisances); full table in Appendix F.

City	screened	$\hat{\theta}$ [90% CI]	b^* (SD) [90% CI]	$b^*/ \hat{c}_Y $
Delhi	no	0.37 [0.29, 0.44]	82.3 [54.3, 144]	560
Mumbai	no	2.23 [-0.66, 4.89]	11.9 [1.3, 43]	392
Bengaluru	no	0.01 [-0.11, 0.14]	26.8 [3.0, 445]	158
Kolkata	no	1.79 [-0.74, 4.60]	10.9 [1.3, 105]	35
Mumbai	yes	0.31 [-2.80, 2.88]	1.2 [0.5, 18]	39
Bengaluru	yes	0.02 [-0.12, 0.16]	35.3 [3.3, 375]	189
Kolkata	yes	4.53 [1.48, 7.42]	16.5 [4.2, 60]	38

values (a nearly constant sensor): its screened estimate 4.5 (1.5–7.4) should not be trusted. Mumbai’s HMM likelihood is multimodal (App. A); using the best of eight converged EM restarts changes its b^* from 11.9 to 7.1 SD without changing any conclusion. The bootstrap resamples fixed cross-fitted residuals and does not refit the HMM or the nuisances, and the point estimate of b^* lies outside its percentile interval in the 2-hourly Bengaluru (screened) and Kolkata (unscreened) cases; b^* is uninformative where $\hat{\theta} \approx 0$. The HMM in the real-data analysis uses diagonal covariances, so the full-covariance Mahalanobis d^2 of Sec. 7 applies only to the simulations. The posterior is fitted on rows that are complete cases of the hourly grid (about 97–98% of consecutive rows are exactly one hour apart; the rest follow gaps).

8 DISCUSSION AND LIMITATIONS

We gave a closed-form law, with every input estimable except one parameter (and the plug-in $\hat{v}$ is only a lower bound when the emission model is misspecified), for the bias of soft conditioning on a latent Markov posterior. It explains when posterior-based DML fails, why a standard conditioning diagnostic is not monotone in reliability, and how to bound the damage. **Practical guidance.** (a) Report $\hat{v}$, $\hat{a}$ and the robustness value alongside every posterior-conditioned effect. (b) Do not read $\lambda_{\min}$ of the residualized Gram matrix as a reliability index when a latent confounder may remain; fit nuisances on $(X, \hat{\gamma})$. (c) Prefer soft posteriors to

hard labels, recalibrate over-confident posteriors, and always compare with regime-fixed-effects and proxy-as-control baselines. **Limitations.** (1) The law is a probability-limit identity under Assumptions 1–3; it is not a finite-sample or uniform-over-HMM-error guarantee, and Assumption 2 (locality of the posterior) is asserted via filter stability, and the $N^{-1/2}$ rate for learned parameters (Prop. 5) rests on hypotheses H1–H3 that we do not verify for EM. (2) Assumption 1 (treatment noise independent of the proxies) is strong; correlation between the NO_2 innovations and meteorology would add terms. (3) The model is linear in the regime indicators and has a single scalar treatment. (4) b is unidentified, so the sensitivity analysis bounds rather than corrects. (5) The plug-in $\hat{v}$ requires calibration and is defined relative to the nuisance learner class: in a test where X_1 informs the regime, the Z -only $\hat{v}$ predicted the bias within 3% (0.30 vs. 0.31), whereas an HMM refit on (Z, X) gave a smaller $\hat{v}$ and underpredicted it (0.24), because a linear nuisance cannot exploit the information the HMM uses. (6) The real-data analysis is illustrative: the proxies partly overlap the controls, lagged pollutants may be mediators, and data-quality issues are material. (7) v depends on the nuisance learner class. Boosted nuisances with the smoother: 4.3% median error for $\text{law}(\hat{v})$ and 14.2% for $\text{law}(v_N)$; calibration experiment (E4): $\text{law}(v_N)$ overshoots by 26%; Table-1 DGP (E3): $\text{law}(\hat{v})$ is 9–32% below the excess bias. (8) The a priori formulas (Wiener, change-point) are approximations with fitted constants; the change-point derivation is a sketch. (9) The formula is an OVB identity; the new content is its calibrated, dependent-data form and its consequences for DML diagnostics.

AI Use Statement

We used large language models to assist with code, derivations, and drafting. The authors reviewed the arguments; experiments are scripted and the numbers are checked against `reports/` and `reports/bias_law/` by `verify_science.py` and `verify_artifacts.py`, though some quoted values are summary roundings and a few derivation steps are explicitly marked as sketches. The authors take full responsibility for the content.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the algorithm, mapping, or model. **[Yes]** Section 2, Section 4.
 - (b) An analysis of the properties and complexity. **[Yes]** Theorem 1, Corollary 1, Propositions 1–2.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **[Yes]** Supplementary material (`src/bias_law/`, `requirements.txt`).
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Assumptions 1–3; strong ones are listed in the limitations.
 - (b) Complete proofs of all theoretical results. **[Partial]** Appendix A gives the algebra in full; the mixing/cross-fitting steps follow standard arguments and are given in outline.
 - (c) Clear explanations of any assumptions. **[Yes]** Sections 3 and 7.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** `README.md`; `python src/bias_law/run_experiments.py`.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Section 7, Appendix D.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., confidence intervals, p-values). **[Yes]** Replication counts and bands are stated; real-data intervals are block bootstrap.
 - (d) A description of the computing infrastructure used. **[Yes]** Appendix D (2-CPU container; most scripts take minutes, the change-point and rate checks up to about an hour).
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
 - (a) If your work uses existing assets, did you cite the creators? **[Yes]** Data sources are described in Appendix F.
 - (b) Did you mention the license of the assets? **[Not Applicable]** Public environmental measurements.
 - (c) Did you include any new assets either in the supplemental material or as a URL? **[Yes]** Cleaned data and code in the supplement.
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? **[Not Applicable]**
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **[Not Applicable]**
5. If you used crowdsourcing or conducted research with human subjects: **[Not Applicable]**

A PROOFS

Proof of Theorem 1. *Step 0 (exact decomposition).* By Assumption 1, V_t is independent of $\mathcal{G} = \sigma(S, Z, X)$, which contains F_t and D_t , so $\mathbb{E}[T_t | F_t] = m(X_t) + a^\top c_t$ and $\mathbb{E}[Y_t | F_t] = \theta \mathbb{E}[T_t | F_t] + g(X_t) + b^\top c_t$. With nuisance errors $\delta_t^T = \hat{m}_T(F_t) - \mathbb{E}[T_t | F_t]$, $\delta_t^Y = \hat{m}_Y(F_t) - \mathbb{E}[Y_t | F_t]$ and $\delta^h = \delta^Y - \theta \delta^T$,

$$\tilde{T}_t = a^\top \tilde{D}_t + V_t - \delta_t^T, \quad \tilde{Y}_t - \theta \tilde{T}_t = b^\top \tilde{D}_t + U_t - \delta_t^h,$$

and $\hat{\theta} - \theta = \sum_t \tilde{T}_t (\tilde{Y}_t - \theta \tilde{T}_t) / \sum_t \tilde{T}_t^2$ exactly.

Step 1 (main terms). $N^{-1} \sum (a^\top \tilde{D}_t)(b^\top \tilde{D}_t) = a^\top \Sigma_N b$ and $N^{-1} \sum (a^\top \tilde{D}_t)^2 = a^\top \Sigma_N a$. Also $N^{-1} \sum V_t^2 = \sigma^2 + O_P(N^{-1/2})$ by the summable-autocovariance condition.

Step 2 (noise cross terms). Conditionally on $\mathcal{G}$, $\sum_t V_t b^\top \tilde{D}_t$ and $\sum_t a^\top \tilde{D}_t U_t$ have mean zero (Assumption 1) and variance at most $C \sum_t \|\tilde{D}_t\|^2 \leq CN$ by summable autocovariances; similarly $\sum_t V_t U_t$ (conditioning on $\mathcal{G}$ and V). Hence each is $O_P(N^{1/2})$.

Step 3 (nuisance terms). Within a cross-fitting fold, $\delta_t = \delta(F_t)$ is a fixed function given the training folds, and $\mathbb{E}[\tilde{D}_t | F_t] = 0$. The sums $\sum \delta_t \tilde{D}_t$, $\sum \delta_t V_t$, $\sum \delta_t U_t$ have conditional mean zero up to an $O(\varepsilon_\tau)$ dependence between training and test blocks (embargo plus Assumption 2) and variance $\leq CN \|\delta\|_{L^2}^2$ by the mixing covariance inequality (Davydov), so they are $O_P(rN^{1/2})$. The product term satisfies $|\sum \delta_t^T \delta_t^h| \leq Nr_T r_h$ by Cauchy–Schwarz.

Step 4. Write the ratio as $(A + e_1)/(B + e_2)$ with $A = a^\top \Sigma_N b$, $B = a^\top \Sigma_N a + \sigma^2 \geq \sigma^2$, and $e_1, e_2 = O_P(N^{-1/2} + r_T r_h + \varepsilon_\tau)$. Then $(A + e_1)/(B + e_2) - A/B = (e_1 - (A/B)e_2)/(B + e_2)$ and $|A/B| \leq \sqrt{b^\top \Sigma_N b}/(2\sigma)$ is bounded (Corollary 1(i), matrix form via Cauchy–Schwarz and $x/(x^2 + \sigma^2) \leq 1/(2\sigma)$). $\square$

Proof of Corollary 1. (i) $a^2 v + \sigma^2 \geq 2|a|\sqrt{v}\sigma$ (AM–GM), with equality at $|a| = \sigma/\sqrt{v}$. (ii) $|a| = \sqrt{(\lambda - \sigma^2)/v}$ gives $|B| = |b|v|a|/\lambda$; and $\frac{d}{d\lambda} \sqrt{\lambda - \sigma^2}/\lambda = (\sigma^2 - \lambda)/(2\lambda^2 \sqrt{\lambda - \sigma^2})$. (iii) $\partial_v[v/(a^2 v + \sigma^2)] = \sigma^2/(a^2 v + \sigma^2)^2 > 0$. $\square$

Proof of Proposition 1. For $\mathcal{F} \subseteq \mathcal{F}'$ the law of total variance gives $\mathbb{E}\text{Cov}(D | \mathcal{F}') \preceq \mathbb{E}\text{Cov}(D | \mathcal{F})$; hard labels are a measurable function of $\hat{\gamma}$, hence $\sigma(\text{hard}) \subseteq \sigma(\hat{\gamma})$. Part (iii) of Corollary 1 finishes the $K = 2$ case. $\square$

Proof of Proposition 2. With $c = \mathbb{E}[D | F]$, $\gamma(1 - \gamma) - c(1 - c) = (\gamma - c)(1 - 2c) - (\gamma - c)^2$, so $|\mathbb{E}[\gamma(1 - \gamma) - c(1 - c)]| \leq \mathbb{E}|\gamma - c| + \mathbb{E}(\gamma - c)^2 \leq \sqrt{CE} + CE$ (Cauchy–Schwarz, $|1 - 2c| \leq 1$), and $\mathbb{E}c(1 - c) = \mathbb{E}\text{Var}(D | F) = v$. If m is linear and $c = \hat{\gamma}$, then $\mathbb{E}[T | F] = m(X) + a^\top \hat{\gamma}$ is linear in $(X, \hat{\gamma})$ and the OLS coefficient is a ; $\text{Var}(\tilde{T}) = a^\top \Sigma a + \sigma^2$ is Step 1. $\square$

Learned HMM parameters. Let $\gamma_{t,\vartheta}$ be the smoothing posterior under parameters ϑ and $\gamma_t^* = \gamma_{t,\vartheta^*}$ with ϑ^* the true parameter. We use three hypotheses, none of which we verify for EM as implemented (local optima, label switching): (H1) the emission model is correct, ϑ^* is interior and identifiable, and the fitted $\hat{\vartheta}$ is consistent (Leroux, 1992; Douc et al., 2011) and $\sqrt{N}$ -asymptotically normal, so $\|\hat{\vartheta} - \vartheta^*\| = O_P(N^{-1/2})$ (Bickel et al., 1998) (also assumed for the EM solution); (H2) $\sup_{\|\vartheta - \vartheta^*\| \leq r} \|\partial_\vartheta \gamma_{t,\vartheta}\| \leq C$ almost surely, uniformly in t, N (forgetting of the smoother and its derivative under strong mixing; Cappé et al., 2005); (H3) X carries no information on S beyond $Z_{1:N}$: $\mathbb{E}[D_t | Z_{1:N}, X] = \mathbb{E}[D_t | Z_{1:N}]$.

Proposition 5 (Rate for learned parameters, smoother). *Let $\hat{\gamma}_t = \gamma_{t,\hat{\vartheta}}$ be the smoothing posterior and assume (H1)–(H3) with (H2) strengthened to $\|\partial_\vartheta \gamma_{t,\vartheta}\| \leq C$ almost surely on a neighbourhood of ϑ^* . Then $\mathbb{E}\|\hat{\gamma}_t - \gamma_t^*\|^2 \leq C' \mathbb{E}\|\hat{\vartheta} - \vartheta^*\|^2 = O(N^{-1})$ uniformly in t (so $\hat{\gamma}_t - \gamma_t^* = O_P(N^{-1/2})$). Write $v^* = \mathbb{E}\gamma^*(1 - \gamma^*) = \mathbb{E}\text{Var}(D | Z_{1:N})$ and let $v_N = \mathbb{E}\text{Var}(D | X, \hat{\gamma})$ be the unrestricted conditional variance. Then in expectation*

$$|\mathbb{E}\hat{v} - v^*| \leq \mathbb{E}|\hat{\gamma} - \gamma^*|, \quad 0 \leq v_N - v^* \leq \mathbb{E}(\hat{\gamma} - \gamma^*)^2, \quad CE := \mathbb{E}(\hat{\gamma} - \mathbb{E}[D | X, \hat{\gamma}])^2 \leq \mathbb{E}(\hat{\gamma} - \gamma^*)^2,$$

so $|\mathbb{E}\hat{v} - v_N| \leq \sqrt{CE} + CE$ and all gaps are $O(N^{-1/2})$ ($v_N - v^*$ and CE are $O(N^{-1})$). The proposition concerns the unrestricted conditional variance; it does not cover restricted learners (linear, boosted), for which v_N differs from it at order one (e.g. $\hat{v} = 0.063$, linear 0.070, boosted 0.079 at $\Delta_Z = 0.5$), nor the filter (which needs (H3) for the filtration $Z_{1:t}$). Passing from these expectations to the realized sample quantity Σ_N of Thm. 1 needs a uniform law of large numbers over a neighbourhood of ϑ^* , which we assume and do not prove.

Proof. The first claim is a mean-value expansion $\hat{\gamma}_t - \gamma_t^* = \partial_{\hat{\gamma}} \gamma_{t,\hat{\gamma}}(\hat{\gamma} - \gamma^*)$ with the almost-sure derivative bound and (H1). Since $|x(1-x) - y(1-y)| \leq |x-y|$, $|\mathbb{E}\hat{v} - v^*| \leq \mathbb{E}|\hat{\gamma} - \gamma^*|$. By (H3), $\mathbb{E}[D_t | X, \hat{\gamma}] = \mathbb{E}[\gamma_t^* | X, \hat{\gamma}] = P\gamma_t^*$, where P is the (non-expansive) L^2 projection onto $\sigma(X, \hat{\gamma})$; as $\hat{\gamma}_t$ is $\sigma(X, \hat{\gamma})$ -measurable, $P\hat{\gamma}_t = \hat{\gamma}_t$ and $CE = \|\hat{\gamma} - P\gamma^*\|^2 = \|P(\hat{\gamma} - \gamma^*)\|^2 \leq \|\hat{\gamma} - \gamma^*\|^2$. Next, $c := \mathbb{E}[D | X, \hat{\gamma}] = P\gamma^*$ has $\mathbb{E}c = \mathbb{E}\gamma^*$, so $v_N - v^* = \mathbb{E}(\gamma^*)^2 - \mathbb{E}c^2 = \|\gamma^* - c\|^2 \geq 0$ (Pythagoras), and $\|\gamma^* - c\| \leq \|\gamma^* - \hat{\gamma}\|$ because c is the best $\sigma(X, \hat{\gamma})$ -measurable approximation of γ^* . Prop. 2 gives $|\mathbb{E}\hat{v} - v_N| \leq \sqrt{CE} + CE$. $\square$

Empirical rate (`x21_hmm_rate.csv`, 24 replications per $N \in \{500, \dots, 16000\}$, $\rho = 0.9$, $\Delta_Z = 0.8$): when the emission model is correct, $\mathbb{E}|\hat{\gamma} - \gamma^*|$ falls from 6.7×10^{-3} to 1.3×10^{-3} with log-log slope -0.49 and the median $|\hat{v} - v^*|$ with slope -0.55 , as H1–H2 predict. With the diagonal-covariance HMM used in our experiments and correlated proxies (misspecified emissions, so H1 fails), the error plateaus (slope -0.09 ; $|\hat{v} - v^*| \approx 2 \times 10^{-3}$, about 9% of v^*): $\hat{\gamma}$ then converges to a pseudo-true posterior, not to the true one. We tested it directly (`x29`): for correlated t_5 proxies ($\Delta_Z=0.5$, $N=3000$, 24 replications) the in-sample plug-in $\hat{v}$ equals the pseudo-true $v(\vartheta_{KL})$ (signed gap -10^{-4} , standard deviation 0.002, maximum 0.005), so `x21`'s plateau is small in this sense, but both understate the realised residual variance $\mathbb{E}(S - \hat{\gamma})^2$ by a factor of two (0.059 versus 0.122). The law with the realised variance predicts the bias (0.326 versus observed 0.308), whereas the plug-in law predicts 0.168. With well-specified Gaussian emissions the understatement is 14% (0.063 versus 0.073; law 0.204 versus 0.194, plug-in 0.178). Under misspecification the plug-in $\hat{v}$ is therefore a lower bound on the bias-relevant variance, not an estimate, and the calibration bound of Prop. 2 (whose CE needs S) cannot be checked from data. *Checks of H1–H3 for the fit we actually use* (`x24–x27`). The HMM is a NumPy EM capped at 25 iterations from a fixed quantile initialisation (`hmmlearn` is not installed). In the simulation designs ($N=3000$, 8 replications for each $\Delta_Z \in \{0.5, 0.8, 1.2\}$) it is indistinguishable from the converged fit (log-likelihood gap $\leq 4 \times 10^{-4}$, identical $\hat{v}$ to four digits), and five random restarts reach the same optimum within 6×10^{-6} (H1 holds in practice). The finite-difference sensitivity $\partial \gamma_t / \partial \vartheta$ at the truth has root-mean-square 0.03–0.13 per parameter, flat in $N \in [10^3, 6.4 \times 10^4]$, while its maximum over t is 0.3–1.8 and drifts up slowly with N : consistent with H2 but not a proof of an almost-sure bound. On the real series, Delhi, Bengaluru and Kolkata have a single optimum, but Mumbai's likelihood is multimodal: five restarts span 1,182 log-likelihood units and the default fit is 118 below the best restart, so H1 fails there; with the best of eight converged restarts Mumbai's $\hat{v}$ moves from 0.021 to 0.025 and b^* from 11.9 to 7.1 SD, and no conclusion changes. H3 fails on the real series: the fitted posterior is predictable from the controls ($R^2 = 0.74/0.73/0.75$ with ridge and 0.96/0.94/0.89 with boosting for Delhi/Bengaluru/Kolkata, 0.19/0.28 for Mumbai) because the controls contain the proxy variables, so $\hat{v}$ there overstates the variance left given $(X, \hat{\gamma})$, and the robustness values are conservative with respect to this channel but not otherwise validated.

Cross-fitting when the HMM is fitted on all of Z . The posterior $\hat{\gamma}_t$ is $\sigma(Z, X)$ -measurable, so $\mathcal{G} = \sigma(S, Z, X)$ contains every input to $\hat{\gamma}$ and the nuisance features. This gives a partial argument, under the additional assumption that (V, U) is α -mixing jointly with $\mathcal{G}$ (stronger than Assumption 1, and violated when controls include lagged outcomes, as in the real data): (i) *Noise terms.* A learner trained on other folds is a function of $\mathcal{G}$ and the training noise, so for a test index t the conditional means of V_t and U_t given $\mathcal{G}$ and the training noise vanish exactly if the noise is independent across folds and up to $\alpha(\tau)$ otherwise. Fitting the HMM on held-out Z adds nothing here because the noise is independent of Z . (ii) *Regime terms.* Given $(Z_{1:N}, X)$, S is an inhomogeneous Markov chain whose dependence decays geometrically under strong mixing of the transition kernel (forgetting of the conditional chain, Cappé et al., 2005), so covariances between training-fold functions of S_s and the test-fold S_t given (Z, X) are $O(\varrho^{|t-s|})$. The test residual $\tilde{D}_t$ is centred given F_t , not given (Z, X) : $\mathbb{E}[\tilde{D}_t | Z, X] = \gamma_t^* - c_t$, which is $O_P(N^{-1/2})$ under H1–H3, so the regime cross terms are $O(\varrho^\tau) + O_P(N^{-1/2})$. The global dependence of $\hat{v}$ on test-fold Z adds a further $O_P(N^{-1/2})$ coupling that we do not bound separately. Empirically (`x22_cond_decay.csv`; known parameters, 40 posterior draws of S given Z , $N=20000$), the correlation of $S_t - \gamma_t$ and $S_{t+h} - \gamma_{t+h}$ decays geometrically, from 0.29–0.77 at $h = 1$ to below 0.006 at $h = 12$ for $\rho = 0.9$ (0.054 at $\rho = 0.97$, $\Delta_Z = 0.5$) and below 0.003 at $h = 24$ in all six settings ($\Delta_Z \in \{0.5, 0.8, 1.2\}$, $\rho \in \{0.9, 0.97\}$). The embargo of 24 used throughout is adequate in these designs; ϱ is not bounded in general (a persistent chain with weak proxies has ϱ near 1).

Direct test (`x30`): refitting the HMM on the training folds only (purged blocks, $\tau=24$, five folds) rather than on all of Z changes neither the bias nor the plug-in v in 24 replications: the fold-wise and global estimates of $\hat{\theta} - \theta$ have correlation 0.999, mean difference 6×10^{-4} (bias 0.308 versus 0.308), and $\hat{v}$ differs by 2×10^{-4} (well-specified design: correlation 0.998, mean $|\Delta \hat{v}| = 5 \times 10^{-4}$). Cross-fitting the HMM is therefore not what drives the bias in these designs; this is evidence, not a proof of independence.

B HETEROGENEOUS EFFECTS AND MULTIPLE REGIMES

Heterogeneous effects ($K = 2$). Let $Y_t = \theta(S_t)T_t + g(X_t) + bD_t + U_t$ with $\theta(0) = \theta_0, \theta(1) = \theta_1, \bar{\theta} = \mathbb{E}\theta(S)$, $T_t = \bar{m} + m_0(X_t) + aD_t + V_t$ with $\mathbb{E}m_0 = 0$, and assume $\mathbb{E}[m_0(X)\text{Var}(D | F)] = 0$. Then

$$\hat{\theta} \xrightarrow{p} \bar{\theta} + \frac{av[b + a(\theta_1 - \bar{\theta}) + \bar{m}(\theta_1 - \theta_0)]}{a^2v + \sigma^2}.$$

Proof. $\mathbb{E}[\tilde{T}\tilde{Y}] = \mathbb{E}[\tilde{T}\theta(S)T] + av$; expanding with $\tilde{T} = a\tilde{D} + V$ gives $\mathbb{E}[\tilde{T}\theta(S)T] = a\bar{m}(\theta_1 - \theta_0)v + a^2\theta_1v + \sigma^2\bar{\theta}$, using $\mathbb{E}[\tilde{D}\theta(S)] = (\theta_1 - \theta_0)v$ and $\mathbb{E}[V^2\theta(S)] = \sigma^2\bar{\theta}$. Subtract $\bar{\theta}(a^2v + \sigma^2)$. $\square$ The oracle-state estimator ($v = 0$) converges to θ .

$K > 2$. Theorem 1 holds with matrix Σ_N and $|B| \leq \sqrt{b^\top \Sigma b}/(2\sigma)$; the hump persists along any direction $a = sa_0$. Hard-versus-soft monotonicity (Prop. 1) holds in the Loewner order but $|B|$ need not be monotone in Σ for $K > 2$.

C EXTENDED COMPARISON OF ESTIMATORS

Bias (RMSE) of $\hat{\theta} - \theta$ with $\theta=1$ (ATE target for the heterogeneous row), $N=3000$, 12 replications ($\Delta_g=3, \Delta_Z=0.5$, persistence 0.9; continuous rows: 12 replications, AR(1) coefficient 0.97). Posterior-DML uses cross-fitted nuisances on $(X, \hat{\gamma})$. Code: `src/bias_law/ext.py`; raw output `reports/bias_law/x*.csv` (x1–x30; x1–x5 comparisons, x6 Kalman/sieve, x7 real joint, x8–x9 finite-state/X-aware, x10–x11 environments, x12–x20 change-point checks, x21–x22 learned-HMM rate and decorrelation, x23–x27 hourly-grid rerun and EM checks, x28 total change-point constant, x29–x30 misspecified- v and fold-wise HMM). Change-point constant: $vd e^{d^2/8}/\epsilon$ for the stationary exact HMM at $d = 4$ is 1.63, 1.46, 1.30 for $\epsilon = 0.1, 0.03, 0.01$ (x16_hmm_limit.csv; its rows at $\epsilon \leq 0.003$ are Monte Carlo noise and are superseded by the windowed x19_hmm_window.csv), against the nearest-neighbour limit $\sqrt{\pi/2} = 1.25$ (x20_nn_quad.csv); asymmetric chains: x14_asym.csv. With Z uninformative ($\Delta_Z=0$) and $\Delta_g=3$ the joint MLE is still unbiased for every $\Delta_T \in \{0.5, \dots, 4\}$ (bias $|\cdot| \leq 0.02$) because the mixture in (T, Y) is identifiable under Gaussianity; non-identification without proxies therefore holds only without parametric assumptions on the mixture.

Table 7: Estimator comparison under misspecification.

Scenario	Δ_T	Posterior-DML	Joint MLE	Proximal 2SLS
HMM correctly specified	1	+0.20 (0.20)	+0.00 (0.02)	-0.01 (0.02)
HMM correctly specified	4	+0.40 (0.40)	+0.00 (0.02)	+0.02 (0.06)
nonlinear $f(X)$ (GBM nuisances)	1	+0.16 (0.17)	-0.01 (0.01)	-0.00 (0.04)
nonlinear $f(X)$ (GBM nuisances)	4	+0.37 (0.37)	-0.00 (0.01)	-0.03 (0.06)
heavy-tailed (t_3) proxy noise	1	+0.15 (0.16)	+0.01 (0.02)	+0.00 (0.04)
heavy-tailed (t_3) proxy noise	4	+0.32 (0.32)	-0.00 (0.01)	-0.04 (0.05)
true $K=3$, fitted $K=2$	1	+0.46 (0.47)	-0.00 (0.02)	-0.01 (0.05)
true $K=3$, fitted $K=2$	4	+0.56 (0.57)	+0.00 (0.01)	+0.06 (0.16)
heterogeneous $\theta(S)$ (ATE target)	1	+0.26 (0.26)	-0.01 (0.03)	-0.02 (0.06)
heterogeneous $\theta(S)$ (ATE target)	4	+0.66 (0.66)	-0.00 (0.02)	-0.01 (0.08)
negative control leaks ($W \ni 0.3T$)	1	+0.21 (0.21)	+0.01 (0.03)	-0.45 (0.45)
negative control leaks ($W \ni 0.3T$)	4	+0.39 (0.39)	-0.00 (0.01)	-0.46 (0.47)
continuous AR(1) latent, SNR 0.5	1	+0.83 (0.84)	+0.79 (0.80)	-0.02 (0.04)
continuous AR(1) latent, SNR 1	1	+0.77 (0.78)	+0.79 (0.79)	-0.02 (0.04)
continuous AR(1) latent, SNR 0.5	2	+0.90 (0.90)	+0.89 (0.89)	-0.03 (0.08)
continuous AR(1) latent, SNR 1	2	+0.87 (0.88)	+0.88 (0.89)	-0.02 (0.06)

D EXPERIMENTAL DETAILS

Learned-HMM grid. $K = 2$: $\Delta_Z \in \{0.5, 0.8, 1.2\}$, $\Delta_T \in \{0.25, 0.5, 1, 2, 4, 8, 16, 32\}$, 12 replications per cell, $N = 3000$, persistence 0.9, $\Delta_g = 3$, $\sigma = 1$. The HMM is fitted once per replication (Gaussian emissions, diagonal covariance, EM); smoothing and filtering posteriors are both used. Predicted vs. observed peak (value of Δ_T at the maximum measured bias, grid-limited): $\Delta_Z = 0.5$: 4.0 vs. 4.0; $\Delta_Z = 0.8$: 6.7 vs. 8; $\Delta_Z = 1.2$: 14.1 vs. 16 (linear nuisances, smoother); predicted peak heights 0.375/0.224/0.106 vs. observed 0.392/0.226/0.117. `reports/bias_low/e1_peak_check.csv`.

$K = 3$. Proxy means on a circle of radius $1.8\Delta_Z$; $(a, b) \in \{(0, 2, 5; 0, 3, 1), (0, 3, -3; 0, 2, 2), (0, 1, 1; 0, 3, -3), (0, 6, 3; 0, 3, 3)\}$; 20 replications; linear nuisances.

Table-1 DGP. `generate_difficulty_dgp` with $\Delta_T = 4$, outcome offsets 15 and 50, state-free treatment level $\bar{m} = 2$, $\theta = (0.75, 2.5)$; the oracle-state estimator provides $\bar{\theta}$.

Compute. 2-CPU container, single-threaded nuisances; most scripts run in minutes, and the long Monte Carlo checks (`x16`, `x19`, `x21`) take up to about an hour. Seeds are fixed in the scripts.

E COUPLED OR-DML AND THE BENCHMARK

The coupled regime-weighted estimator $\hat{\theta} = (\hat{J} + \lambda I)^{-1} \hat{S}$ of our earlier work, with a spectral penalty $\lambda \propto N^{-1/2}$, is evaluated against honest baselines (Table 8; 100 replications per Δ_Z). At $\Delta_Z \geq 2$ it ties regime fixed effects built from the same HMM; at $\Delta_Z \leq 1$ every method fails, with regime fixed effects better at $\Delta_Z = 1$. The spectral penalty is numerically inert (the “spectral” and “unregularized” rows differ by ≤ 0.003 in mean absolute bias at $\Delta_Z = 2$). Coverage for the regime-FE row uses the blocked-DML standard error and should be read as indicative only. We therefore do not present the coupled estimator as a contribution.

F REAL-DATA DETAILS

Data. CPCB monitors via OpenAQ (timestamps at hh:30Z, floored to the hour and paired with the Open-Meteo reanalysis hour that began 30 minutes earlier) and Open-Meteo, regularised to an hourly grid with forward-filling of at most 2 hours (`ext_real_hourly_fix.py`); complete cases: 9,024 (Delhi), 8,113 (Mumbai), 7,775 (Bengaluru), 2,322 (Kolkata). The legacy pipeline (`data_pipeline_clean.py`) used `round('h')`, which sends every hh:30 to an even hour, so its 14,122 complete cases are a 2-hourly series (lags and rolling means span 2 h and 6 h; consecutive rows are 2 h apart). Treatment: NO_2 ; outcome: $\text{PM}_{2.5}$; proxies Z : temperature, wind speed (km/h), humidity, pressure, hour-of-day sine/cosine; controls X : lagged and rolling pollutants plus the meteorological variables (so Z overlaps X ; the plug-in $\hat{v}$ ignores the information in X and can overstate v). Sensor screening: runs of ≥ 6 identical NO_2 readings are *not* removed in the main analysis; Table 6 reports both. Delhi has none; On the hourly grid, screening removes 1,857 rows in Mumbai, 575 in Bengaluru and 861 in Kolkata, whose NO_2 has 16 distinct values. **Newey–West sensitivity of the regime-specific OR-DML estimates.** Standard errors are *not* stable across lags: for example the Bengaluru regime-2 effect has $p = 0.027$ at lag 12, 0.071 at lag 72 and 0.101 at lag 168; Mumbai’s regime-2 effect has $p = 0.026, 0.051, 0.074$. The regime-specific claims of our earlier work should be read accordingly; these estimates also come from the legacy 2-hourly series.

Table 8: Difficulty-frontier benchmark (100 replications per Δ_Z , $N = 1200$, true $\theta_{ATE} = 1.5278$); `reports/or_dml_benchmark_summary.csv`.

Δ_Z	Estimator	mean bias	median bias	RMSE	cover. SATE
0.2	Oracle DML	0.056	0.044	0.070	96%
	Standard DML	8.456	8.456	8.456	0%
	Block DML	8.490	8.479	8.491	0%
	DML + Z Controls (Ridge)	8.307	8.312	8.308	0%
	DML + Z Controls (GBM)	8.018	8.025	8.019	0%
	Regime FE DML (Hard)	8.277	8.281	8.278	0%
	Unregularized Coupled DML	8.640	8.613	8.641	0%
	Spectral OR-DML (Ours)	8.630	8.602	8.631	0%
	Filtered OR-DML (Ours)	8.606	8.587	8.607	0%
0.5	Oracle DML	0.055	0.044	0.071	92%
	Standard DML	8.429	8.432	8.430	0%
	Block DML	8.467	8.483	8.468	0%
	DML + Z Controls (Ridge)	7.429	7.424	7.430	0%
	DML + Z Controls (GBM)	7.031	7.020	7.034	0%
	Regime FE DML (Hard)	6.593	6.632	6.600	0%
	Unregularized Coupled DML	8.847	8.886	8.851	0%
	Spectral OR-DML (Ours)	8.835	8.874	8.839	0%
	Filtered OR-DML (Ours)	8.988	9.001	8.991	0%
1.0	Oracle DML	0.057	0.043	0.073	92%
	Standard DML	8.433	8.449	8.434	0%
	Block DML	8.466	8.475	8.467	0%
	DML + Z Controls (Ridge)	5.438	5.462	5.440	0%
	DML + Z Controls (GBM)	3.633	3.642	3.647	0%
	Regime FE DML (Hard)	2.642	2.689	2.679	0%
	Unregularized Coupled DML	4.025	4.030	4.045	0%
	Spectral OR-DML (Ours)	4.017	4.022	4.037	0%
	Filtered OR-DML (Ours)	5.172	5.185	5.185	0%
2.0	Oracle DML	0.052	0.042	0.066	93%
	Standard DML	8.432	8.451	8.433	0%
	Block DML	8.461	8.472	8.462	0%
	DML + Z Controls (Ridge)	2.631	2.614	2.635	0%
	DML + Z Controls (GBM)	0.207	0.194	0.243	72%
	Regime FE DML (Hard)	0.077	0.063	0.102	96%
	Unregularized Coupled DML	0.082	0.073	0.105	93%
	Spectral OR-DML (Ours)	0.081	0.072	0.104	92%
	Filtered OR-DML (Ours)	0.102	0.081	0.131	92%
4.0	Oracle DML	0.054	0.048	0.067	96%
	Standard DML	8.451	8.461	8.452	0%
	Block DML	8.486	8.495	8.487	0%
	DML + Z Controls (Ridge)	0.848	0.842	0.853	0%
	DML + Z Controls (GBM)	0.101	0.082	0.125	89%
	Regime FE DML (Hard)	0.054	0.046	0.067	100%
	Unregularized Coupled DML	0.054	0.048	0.067	96%
	Spectral OR-DML (Ours)	0.054	0.046	0.067	96%
	Filtered OR-DML (Ours)	0.054	0.046	0.067	96%

Table 9: Full real-data sensitivity table (hourly grid); `reports/bias_low/x23_real_hourly_fix.csv`.

City	screen	N	$\hat{\theta}$ [90% CI]	$\hat{v}$	$\hat{a}$	$\text{Var}(\tilde{T})$	b^* (SD of $\tilde{Y}$)	[90% CI]	$b^*/ \hat{c}_Y $
Delhi	no	9024	0.370 [0.292, 0.436]	0.007	-2.892	134.53	82.30	[54.29, 143.86]	560
Mumbai	no	8113	2.235 [-0.659, 4.890]	0.021	0.036	0.20	11.91	[1.33, 42.63]	392
Bengaluru	no	7775	0.013 [-0.105, 0.135]	0.021	-0.028	6.05	26.76	[3.02, 445.24]	158
Kolkata	no	2322	1.790 [-0.740, 4.602]	0.021	0.016	0.00	10.89	[1.26, 105.13]	35
Delhi	yes	9024	0.370 [0.292, 0.436]	0.007	-2.892	134.53	82.30	[54.29, 143.86]	560
Mumbai	yes	6256	0.308 [-2.796, 2.884]	0.021	0.057	0.25	1.23	[0.45, 17.98]	39
Bengaluru	yes	7200	0.023 [-0.124, 0.157]	0.021	0.034	5.74	35.28	[3.32, 374.91]	189
Kolkata	yes	1461	4.526 [1.484, 7.420]	0.020	0.034	0.00	16.47	[4.18, 59.99]	38

Table 10: Legacy 2-hourly series (pipeline artifact; for comparison only); reports/bias_low/e5_real_city_sensitivity.csv.

City	screen	N	$\hat{\theta}$ [90% CI]	$\hat{v}$	$\hat{a}$	$\text{Var}(\tilde{T})$	b^* (SD of $\tilde{Y}$) [90% CI]	$b^*/ \hat{c}_Y $
Delhi	no	4640	0.584 [0.449, 0.682]	0.005	-4.399	278.54	175.02 [85.37, 604.64]	1228
Mumbai	no	4180	1.903 [-1.672, 5.014]	0.035	-0.033	0.69	21.96 [2.23, 227.45]	366
Bengaluru	no	4093	-0.006 [-0.148, 0.135]	0.025	-0.355	12.21	1.43 [0.54, 109.33]	5
Kolkata	no	1209	0.117 [-2.572, 3.329]	0.026	0.029	0.01	0.44 [0.47, 39.51]	1
Delhi	yes	4640	0.584 [0.449, 0.682]	0.005	-4.399	278.54	175.02 [85.37, 604.64]	1228
Mumbai	yes	3295	4.893 [-0.123, 13.744]	0.017	0.130	0.74	31.01 [2.76, 98.92]	417
Bengaluru	yes	3930	-0.000 [-0.174, 0.128]	0.024	-0.203	12.45	0.20 [2.18, 357.38]	1
Kolkata	yes	1099	0.415 [-2.704, 3.516]	0.024	0.034	0.01	1.49 [0.46, 51.03]	3

Table 11: HAC-lag sensitivity (hours); reports/bias_low/e6_hac_lag_sensitivity.csv.

City	regime	$\hat{\theta}$	SE (12)	SE (24)	SE (72)	SE (168)
Delhi	1	0.704	0.051	0.055	0.063	0.073
Delhi	2	0.128	0.074	0.083	0.109	0.125
Mumbai	1	-0.030	8.716	6.437	4.076	3.053
Mumbai	2	3.303	1.480	1.538	1.692	1.847
Bengaluru	1	0.092	0.155	0.167	0.182	0.167
Bengaluru	2	-0.153	0.069	0.072	0.084	0.093
Kolkata	1	0.679	1.404	1.344	1.338	1.201
Kolkata	2	-1.065	2.668	2.757	3.082	3.346